

#14Days14Chapters

DAY 9

[Aim: 100/100 in Maths]

REAL NUMBERS

Ch-1



HCF & LCM

→ ~~basic Q.~~

→ ~~Word Problems~~

Irrationality

$\sqrt{2}, \sqrt{5}, \sqrt{7} \dots$

#LCM (Least Common Multiple):

$$\begin{array}{l} 6 \\ 8 \end{array} \rightarrow \boxed{\text{LCM} = 24}$$

वो सबसे
छोटा no.
जो दोनों से
divisible हो !!

6 से divide हो जायेगी

$$6 = 6, 12, 18, 24, 30, 36, 42, 48, \dots$$
$$8 = 8, 16, 24, 32, 40, 48, \dots$$

8 से divide हो जायेगी

#HCF (Highest Common Factor):

$$\begin{array}{l} 6 \\ 8 \end{array} \rightarrow \boxed{\text{HCF} = 2}$$

HCF → वो सबसे
बड़ा No. जो
दोनों को divide
कर दे!



जो 6
से
divide
कर देगे



जो 8
से
divide
कर देगे

#LP: Find HCF and LCM of 6 & 8.

$$6 = 2 \times 3 \Rightarrow \underline{2^1} \times \underline{3^1}$$

$$8 = 2 \times 2 \times 2 \Rightarrow 2^3$$

$$\text{HCF} = \text{common factor} \Rightarrow 2$$

$$\text{LCM} = 2^3 \times 3^1 \Rightarrow \underline{24}$$

$$\begin{array}{r|l} 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

#Short Trick:

#LP: Find HCF and LCM of 72 and 120

$$72 = 2 \times 2 \times 2 \times 3 \times 3 \Rightarrow 2^3 \times 3^2$$

$$120 = 2 \times 2 \times 2 \times 3 \times 5 \Rightarrow 2^3 \times 3^1 \times 5^1$$

$$\text{HCF} = 2 \times 2 \times 2 \times 3 \Rightarrow \underline{\underline{24}}$$

$$\text{LCM} = \underline{2^3 \times 3^2 \times 5^1}$$

2	120
2	60
2	30
3	10
5	2
	1

2	72
2	36
2	18
3	9
3	3
	1

#LP: Find the HCF of 612 and 1314 using prime factorisation. ↪ multiplication

$$\begin{array}{r|l} 2 & 1314 \\ \hline 3 & 657 \\ \hline 3 & 219 \\ \hline 73 & 73 \\ \hline & 1 \end{array}$$

$$612 = \underline{2} \times \underline{2} \times \underline{3} \times \underline{3} \times \underline{17}$$
$$1314 = \underline{2} \times \underline{3} \times \underline{3} \times \underline{73}$$

$$HCF = 2 \times 3 \times 3$$

$$\boxed{HCF = 18}$$

[CBSE 2019]

$$\begin{array}{r|l} 2 & 612 \\ \hline 2 & 306 \\ \hline 3 & 153 \\ \hline 3 & 51 \\ \hline 17 & 17 \\ \hline & 1 \end{array}$$

#LP:

Write the smallest number which is divisible by both 306 and 657.

[CBSE 2019]

LCM $\rightarrow$ वो सबसे छोटा No. जो दोनों से divide हो जाए।

LCM $\Rightarrow$

$$306 =$$

$$657 =$$

#LP: If HCF of 65 and 117 is expressible in the form $65n-117$, then find the value of n . (2019)

$$65 = 5 \times 13$$

$$117 = 3 \times 3 \times 13$$

$$\text{HCF} = 65n - 117$$

$$13 = 65n - 117$$

$$13 + 117 = 65n$$

$$2 \cancel{130} = \cancel{65}n \Rightarrow n = 2$$

3		117		5		65
3		39		13		13
13		13				1
		1				

#LP:

If $x = ab^3$ and $y = a^3b$, where a and b are prime numbers, then $[HCF(x, y) - LCM(x, y)]$ is equal to

[CBSE 2025]

- (a) $1 - a^3b^3$
- (b) $ab(1 - ab)$
- (c) $ab - a^4b^4$
- (d) $ab(1 - ab)(1 + ab)$

$$x = a^1 b^3 \Rightarrow a \times b \times b \times b$$

$$y = a^3 b^1 \Rightarrow a \times a \times a \times b$$

$$HCF = ab$$

$$LCM = a^3 b^3$$

$$[HCF - LCM]$$

$$\Rightarrow ab - a^3 b^3$$

$$\Rightarrow ab(1 - a^2 b^2)$$

$$\Rightarrow ab(1^2 - (ab)^2)$$

$$A^2 - B^2 \Rightarrow (A+B)(A-B)$$

$$\Rightarrow \underline{\underline{ab(1+ab)(1-ab)}}$$

#LP:

If $a = 2^2 \times 3^x$, $b = 2^2 \times 3 \times 5$, $c = 2^2 \times 3 \times 7$, and $\text{LCM}(a, b, c) = 3780$, then x is equal to

(a) 1

☒ (b) 2

☐ (c) 3

(d) 0

[CBSE 2024]

$$a = 2^2 \times 3^x$$

$$b = 2^2 \times 3^1 \times 5^1$$

$$c = 2^2 \times 3^1 \times 7^1$$

$$\text{LCM} = 2^2 \times 3^x \times 5^1 \times 7^1$$

$$\text{LCM} = 3780$$

$$2^2 \times 3^x \times 5^1 \times 7^1 = 2^2 \times 3^3 \times 5^1 \times 7^1$$

Comparing $x = 3$

2	3780
2	1890
3	945
3	315
3	105
5	35
7	7
	1

#LP: The ratio of HCF to LCM of the least composite number and the least prime number is

(a) 1:2

(b) 2:1

(c) 1:1

(d) 1:3 (2023)

$$\frac{HCF(4, 2)}{LCM(4, 2)} = \frac{2}{4} \Rightarrow \left(\frac{1}{2}\right) \quad \boxed{1:2}$$

$$4 = 2 \times 2 \Rightarrow 2^2$$

$$2 = 2 \Rightarrow 2^1$$

$$HCF = 2$$

$$LCM = 2^2 = 4$$

Prime no. $\rightarrow$ No. divisible by 1 & itself

eg: 2, 3, 5, 7, 11, 13, 17 - - -

Composite No $\rightarrow$ No. having more than 2 factor

4, 6, 8, 9, 10, 12, 14, 15 - - -

* 1 $\rightarrow$ neither prime nor composite.

∴ 6^n cannot end with 0 because its
prime factorisation doesn't contain
2 & 5 both.

#LP:

If 'n' is any natural number, then $(12)^n$ cannot end with the digit

(a) 2

(b) 4

(c) 8

☒ (d) 0 (Term I, 2021-22)

$$12^n \Rightarrow$$

$$12^1 \Rightarrow 2^2 \times 3 \quad \begin{matrix} 2^{\checkmark} \\ 5 \times \end{matrix}$$

$$12^2 \Rightarrow (2^2 \times 3)^2 \Rightarrow 2^2 \times 3 \times 2^2 \times 3 \quad \begin{matrix} 2^{\checkmark} \\ 5 \times \end{matrix}$$

$$n=n \Rightarrow (2^2 \times 3)^n \Rightarrow 2^2 \times 3 \times 2^2 \times 3 \times 2^2 \times 3 \dots n \Rightarrow \begin{matrix} 2^{\checkmark} \\ 5 \times \end{matrix}$$

123B

$$\frac{6}{2}$$

$$\begin{array}{r} 3 \\ 2 \overline{) 6} \\ \underline{10} \end{array}$$

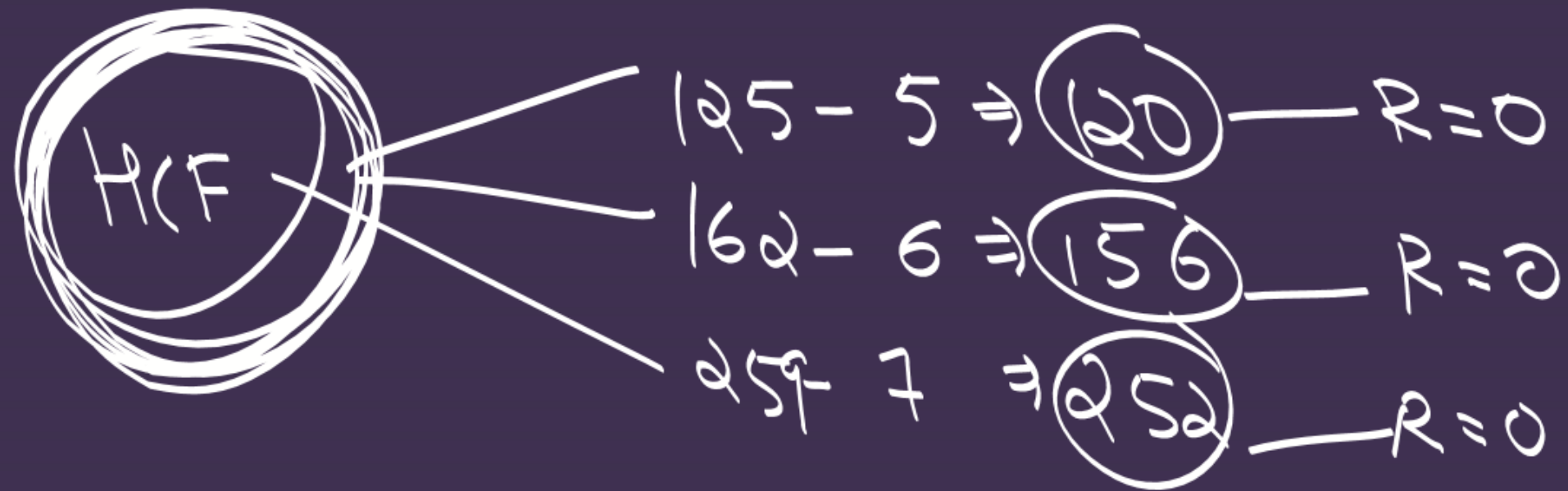
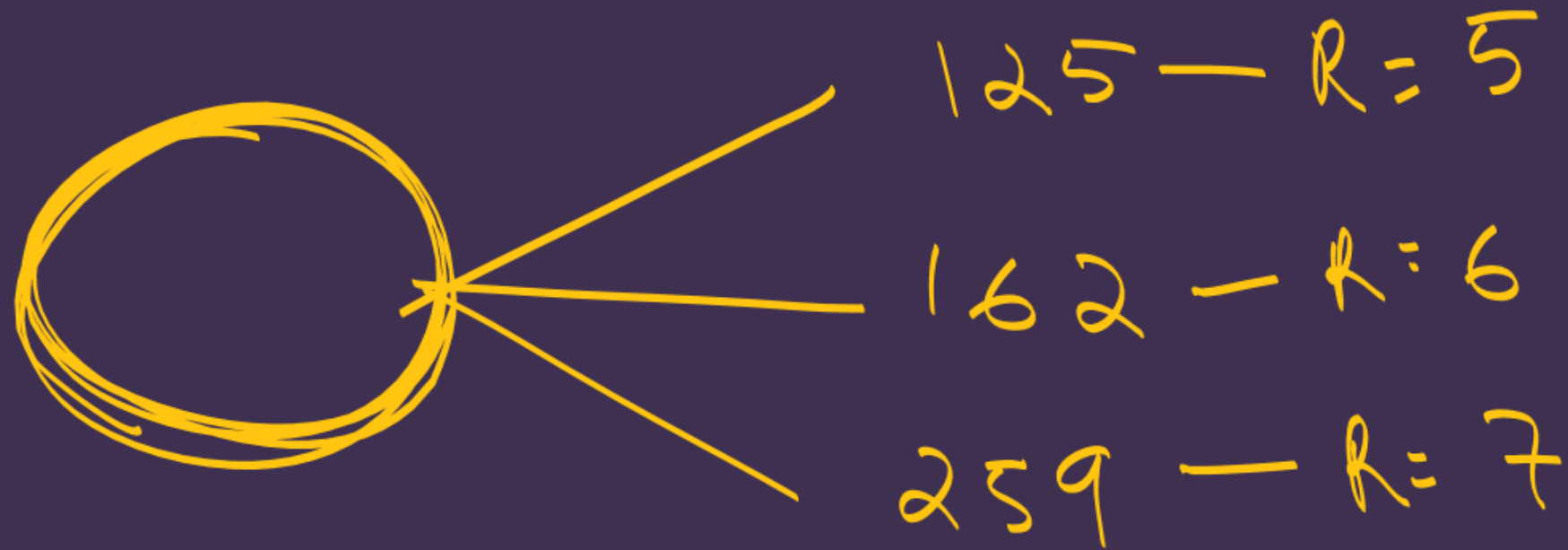
divide ~~हर दिवत~~ $\Rightarrow R=0$

2 is factor of 6

$$\frac{6}{2} \Rightarrow \boxed{R=0}$$

factor हरमा $R=0$

#LP: Find the largest possible positive integer that divides 125, 162 and 259 leaving remainder 5, 6 and 7 respectively. (Board Term 1, 2017)



Final Ans $\rightarrow$ HCF (120, 156, 252)

$$120 =$$

$$156 =$$

$$252 =$$

$$\text{HCF} = 12$$

$$2\sqrt[3]{7}$$

$$R = \frac{1}{1}$$

$$\frac{7}{2}$$

$$7 - \text{[scribble]} \Rightarrow \frac{\text{[scribble]}}{2} \text{ [scribble]}$$

#LP: Find the greatest number which divides 120, 165 and 210 exactly leaving remainders 5, 4 and 3 respectively.

a) 7

b) 5

☒ c) 23

d) None of these



$$HCF(115, 161, 207) = 23$$

#K3B

Prime no.s

prime

$$\begin{array}{l} 2 \\ 5 \end{array} = \begin{array}{l} 2^1 \\ 5^1 \end{array} \times \begin{array}{l} 1 \\ 1 \end{array}$$

$$\text{HCF} = 1$$
$$\text{LCM} = 2^1 \times 5^1$$

prime

$$\begin{array}{l} 3 \\ 7 \end{array} = \begin{array}{l} 3^1 \\ 7^1 \end{array} \times \begin{array}{l} 1 \\ 1 \end{array}$$

$$\text{HCF} = 1$$
$$\text{LCM} = 3^1 \times 7^1$$

✓✓
p & q are two
prime no. $\rightarrow \text{HCF} = 1$ (fix)
 $\rightarrow \text{LCM} = p \times q$

#LP:

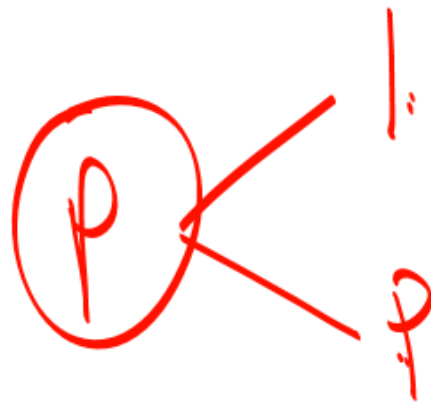
If p is a prime number, what is the sum of its factors?

a) p

b) 1

☒ c) $p+1$

d) $p-1$



#LP:

The LCM of two prime numbers p and q ($p > q$) is 221. Then the value of $3p - q$ is: (2024)

- (a) 4
- (b) 28
- ☒ (c) 38
- (d) 48

$$\text{LCM} = p \times q = 221 \text{ (given)}$$

17	221
13	13
	1

$$p \times q = 13 \times 17$$

$$\because (p > q)$$

∴ comparing $\begin{cases} p = 17 \\ q = 13 \end{cases}$

Final Ans $\Rightarrow 3p - q$

$$\Rightarrow 3(17) - 13$$

$$\Rightarrow 51 - 13 = 38$$

#LP:

If the LCM of two numbers is 3600, then which of the following cannot be their HCF?

- (a) 600 ✓
- (b) 500
- (c) 400 ✓
- (d) 150 ✓

⊛ ⇒ HCF is always factor of LCM.

ie. $HCF \rightarrow LCM$ को divide कर देगा

$$\begin{array}{r} 3600 \\ \hline 500 \end{array}$$

Relation between HCF & LCM:

For two numbers a & b .

$$HCF(a, b) \times LCM(a, b) = a \times b$$

→ $H \times L = \text{Product of two numbers}$

#LP: If the HCF (2520, 6600) = 40 and LCM (2520, 6600) = 252 × k, then the value of k is:

- (A) 1650
- (B) 1600
- (C) 165
- (D) 1625

$$H \times L = 2520 \times 6600$$
$$\cancel{40} \times (\cancel{252} \times k) = \cancel{252}^{10} \times \cancel{6600}^{1650}$$
$$k = 1650$$

#LP: Two numbers are in the ratio $2 : 3$ and their LCM is 180. What is the HCF of these numbers? (2023)

$$\frac{a}{b} = \frac{2}{3}$$

let $\boxed{\begin{matrix} a = 2k \\ b = 3k \end{matrix}}$

$\boxed{\text{HCF} = k}$

$$L \times H = a \times b$$

$\cancel{30} \cancel{90} / \cancel{180} \times \cancel{k} = (\cancel{2} \cancel{k}) (\cancel{3} \cancel{k})$

$\boxed{k = 30}$

$\boxed{\text{HCF} = k = 30}$

#LP: Find the number of possible pairs if the product of two Numbers and HCF are 4500 and 15 Respectively.

$$\text{HCF} = 15$$

Let two no's are

⊛ $15a$ & $15b$

$$\text{Product} = 4500$$
$$\cancel{15a} \times \cancel{15b} = \cancel{4500}^{20}$$

$$a \times b = 20$$

$$2 \times 10 = 20$$

$$4 \times 5 = 20$$

$$20 \times 1 = 20$$

$$(2, 10)$$

$$(4, 5)$$

$$(20, 1)$$

} 3 pairs

FUNDAMENTAL THEOREM OF ARITHMETIC:

According to the fundamental theorem of arithmetic, every composite number can be written (factorise) as the product of primes and this factorisation is unique, apart from the order in which the prime factors occur.

2	12
2	6
3	3
	1

$$\text{12} = 2 \times 2 \times 3$$

Composite

product of prime.

#LP:

Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number. (2024)

$$(let) \ x = 5 \times 11 \times 17 + 3 \times 11$$

$$x = 11(5 \times 17 + 3)$$

$$x = 11 \times 88$$

$$x = \underbrace{11 \times 2 \times 2 \times 2 \times 11}_{\substack{p \quad p \quad p \quad p \quad p}}$$

product of prime

Since it can be written as product of primes, it's a composite
No

$$\begin{array}{r|l} 2 & 88 \\ \hline 2 & 44 \\ \hline 2 & 22 \\ \hline & 11 \\ \hline & 1 \end{array}$$

#LP:

Que : 210. Let x and y be two distinct prime numbers and $p = x^2 y^3$, $q = xy^4$, $r = x^5 y^2$. Find the HCF and LCM of p, q and r. Further check if $\text{HCF}(p, q, r) \times \text{LCM}(p, q, r) = p \times q \times r$ or not.

$$p = x^2 y^3 \Rightarrow x \cdot x \cdot y \cdot y \cdot y$$

$$q = x^1 y^4 \Rightarrow x \cdot y \cdot y \cdot y \cdot y$$

$$r = x^5 y^2 \Rightarrow x \cdot x \cdot x \cdot x \cdot x \cdot y \cdot y$$

$$\text{HCF} = x \cdot y \cdot y \Rightarrow xy^2$$

$$\text{LCM} = x^5 \cdot y^4$$

$$L \times H = p \times q \times r$$

$$\text{LHS} \Rightarrow L \times H \Rightarrow (xy^2)(x^5y^4)$$

$$\text{LHS} \Rightarrow x^6y^6$$

$$\text{RHS} \Rightarrow p \times q \times r$$

$$\Rightarrow x^2y^3 \times xy^4 \times x^5y^2$$

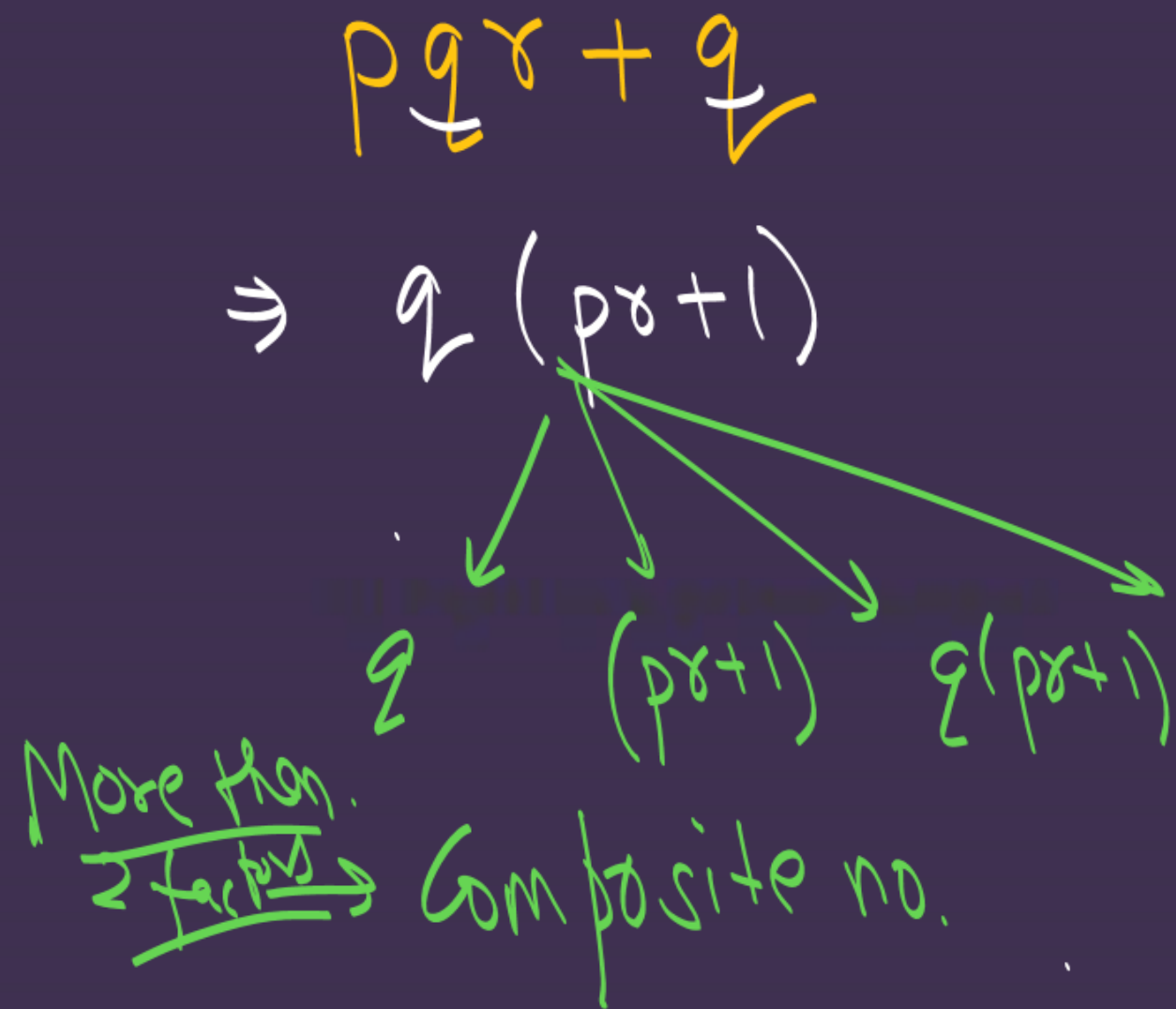
$$\Rightarrow x^8y^9$$

$$\boxed{\text{LHS} \neq \text{RHS}}$$

#LP: Let p , q and r be three distinct prime numbers. Check whether $pqr + q$ is a composite number or not. Further, give an example for three distinct primes p, q, r such that

[CBSE 2025]

- (i) $pqr + 1$ is a composite number
- (ii) $pqr + 1$ is a prime number



(i) $(pqr + 1) \rightarrow \text{comp}$

$3 \ 5 \ 7$

$106 \rightarrow \text{comp}$

$\left. \begin{array}{l} p = 3 \\ q = 5 \\ r = 7 \end{array} \right\} \checkmark$

$(pqr + 1) \rightarrow \text{prime}$

$2 \ 3 \ 5$

$(31) \rightarrow$

$\left. \begin{array}{l} p = 2 \\ q = 3 \\ r = 5 \end{array} \right\} \checkmark$

Tricks to solve Word Problem:

Given में वसति Ans $\rightarrow$ LCM

Given में होता Ans $\rightarrow$ HCF

Word Problems on HCF & LCM:

#LP:

Three bells ring at intervals of 6, 12 and 18 minutes. If all the three bells rang at 6 am., when will they ring together again?

[CBSE 2023]

JEE & NEET

Mission JEE
Mission NEET
Drona JEE
Drona NEET

Next
Toppers
For Sci, Comm, Hum
Class-11th

✓ Prarambh → ~~3499~~

(Early 500) 2999

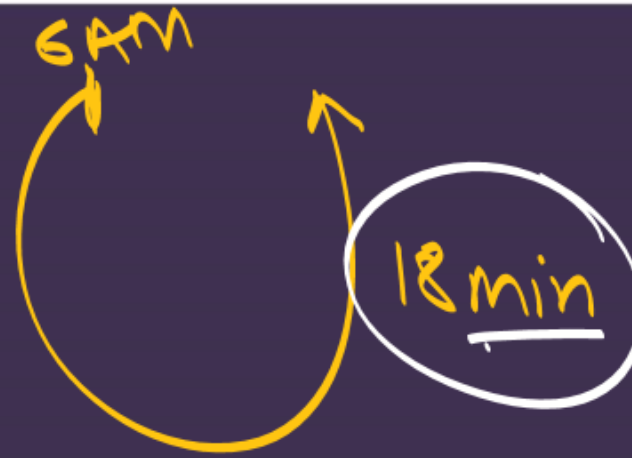
✓ Prarambh Plus → ~~4999~~
(Prarambh + Nirbhay) 3499 (Early 500)

Word Problems on HCF & LCM:

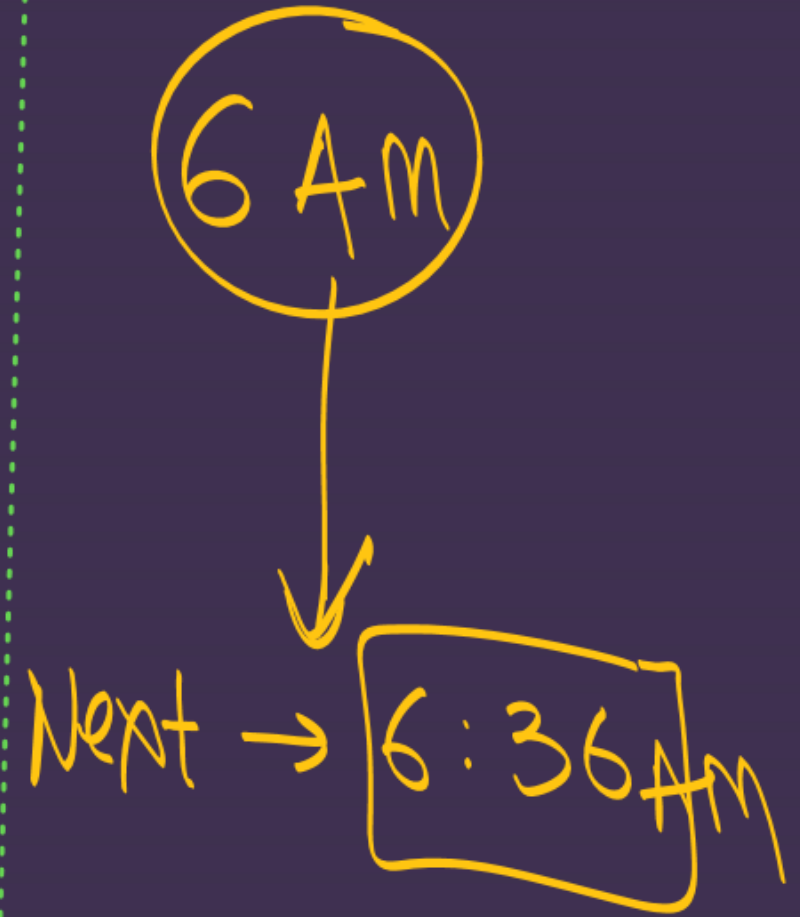
#LP:

Three bells ring at intervals of 6, 12 and 18 minutes. If all the three bells rang at 6 a.m., when will they ring together again?

[CBSE 2023]



Q/S $\rightarrow$ LCM (6, 12, 18) = 36 min



#LP: The traffic lights at 3 different road crossings change after every 48 sec, 72 sec and 108 sec, respectively. If they all change simultaneously at 8:20 a.m., then at what time will they next change again simultaneously? **[CBSE 2020]**

48 sec

72 sec

108 sec

$$\text{Q.S.} \rightarrow \text{LCM}(48, 72, 108) = \underline{\underline{432 \text{ sec}}}$$

$$7 \text{ min} \rightarrow 420 \text{ sec}$$

$$\text{Remain} \rightarrow 12 \text{ sec}$$

8:20 am

8:27:12 ✓

#LP: In a teachers' workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject. (2024)

F $\rightarrow$ 48 teacher

H $\rightarrow$ 80 teacher

E $\rightarrow$ 144 teacher

सोल $\Rightarrow$ HCF (48, 80, 144)

$\downarrow$

$\Rightarrow$ 16 teachers

x x x x x - - -

16 16 16 16 16 - - -

~~Total teachers $\Rightarrow$ 272~~

~~teachers in one room $\rightarrow$ 16~~

~~No. of room $\Rightarrow \frac{272}{16}$~~ 17 room

F

T $\rightarrow$ 48 teachers1 room $\rightarrow$ 16

$\therefore$ No. of rooms for french teachers $\Rightarrow \frac{48}{16} = 3$

F = 3 rooms

$\frac{16}{F} \quad \frac{16}{F} \quad \frac{16}{F} \quad \frac{16}{F}$

H

T $\rightarrow$ 801 room $\rightarrow$ 16

No. of room for hind $\Rightarrow \frac{80}{16} = 5$

H $\Rightarrow$ 5 rooms

E

T $\rightarrow$ 1441 room $\rightarrow$ 16

No. of rooms for E $\Rightarrow \frac{144}{16} = 9$

E = 9 rooms

Q38. February 14 is celebrated as International Book Giving Day and many countries in the world celebrate this day. Some people in India also started celebrating this day and donated the following number of books of various subjects to a public library:

History = 96, science = 240, Mathematics = 336

Ans → H → 2
S → 5
M → 7
stacks

These books have to be arranged in minimum number of stacks such that each stack contains books of only one subject and the number of books on each stack is the same.

Based on the above information, answer the following questions.

- (i) How many books are arranged in each stack? 48 books
- (ii) How many stacks are used to arrange all the Mathematics books? 7 stacks
- (iii) Determine the total number of stacks that will be used for arranging all the books. → $2 + 5 + 7 = 14$ stacks

OR

- (iii) If the thickness of each book of History, Science and Mathematics is 1.8 cm, 2.2 cm and 2.5 cm respectively, then find the height of each stack of history, science and mathematics books.

H $\rightarrow$ 96 books

S $\rightarrow$ 240 books

M $\rightarrow$ 336 books

LCM $\rightarrow$ HCF(96, 240, 336)

each stack $\Rightarrow$ 48 books
has

x x x x - - -
48 books 48 books 48 48 - - -

H

S

M

T $\rightarrow$ 96 books
1 stack $\rightarrow$ 48 books
 $\therefore$ No. of stacks $\rightarrow \frac{96}{48} = 2$

H $\rightarrow$ 2 stacks

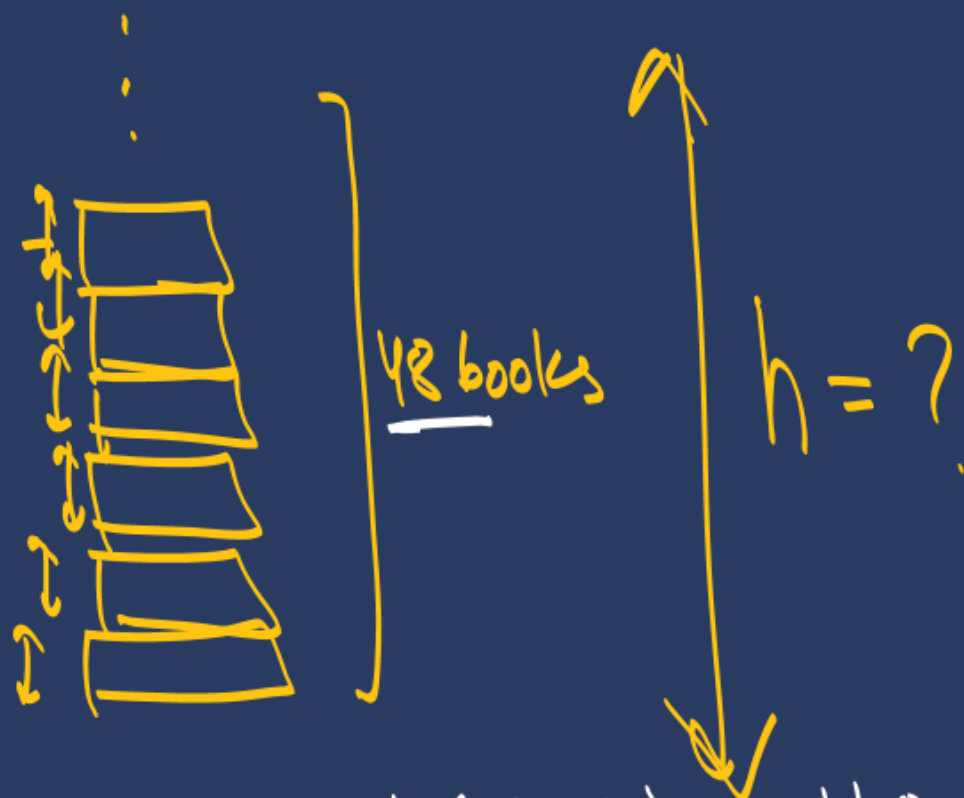
T $\rightarrow$ 240
1 stack $\rightarrow$ 48
No. of stacks $\rightarrow \frac{240}{48} = 5$

S $\rightarrow$ 5 stacks

T $\rightarrow$ 336 books
1 stack $\rightarrow$ 48
No. of stacks $\rightarrow \frac{336}{48} = 7$

M $\rightarrow$ 7 stacks

H



$$\begin{aligned} \text{Total height} &\rightarrow 48 \times 1.8 \\ &\Rightarrow \underline{\underline{86.4 \text{ cm}}} \end{aligned}$$

S



$$\begin{aligned} 1 \text{ book} &\rightarrow 2.2 \text{ cm} \\ 48 \text{ book} &\rightarrow \underline{\underline{(48 \times 2.2) \text{ cm}}} \end{aligned}$$

m



$$\begin{aligned} 1 \text{ book} &\rightarrow 25 \text{ cm} \\ 48 \text{ books} &\rightarrow \underline{\underline{(25 \times 48) \text{ cm}}} \end{aligned}$$

#Irrationality:

#LP: Write whether $\frac{2\sqrt{45} + 3\sqrt{20}}{2\sqrt{5}}$ on simplification gives a rational or an irrational number.

$$\rightarrow \frac{2(3\sqrt{5}) + 3(2\sqrt{5})}{2\sqrt{5}}$$

$$\rightarrow \frac{\cancel{2\sqrt{5}}(3+3)}{\cancel{2\sqrt{5}}} \rightarrow \textcircled{6} \rightarrow \underline{\underline{\text{Rat}}}$$

$$\sqrt{45} = \sqrt{3^2 \times 5}$$

$$\boxed{\sqrt{45} = 3\sqrt{5}}$$

$$\sqrt{20} = \sqrt{2^2 \times 5}$$

$$\boxed{\sqrt{20} = 2\sqrt{5}}$$

$$\begin{array}{r} 3 \overline{) 45} \\ 3 \\ \hline 15 \\ 15 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 20} \\ 2 \\ \hline 0 \end{array}$$

Proof of Irrationality:

#LP: Prove that $\sqrt{2}$ is an irrational number. (2020 C, NCERT, Delhi 2019)

let $\sqrt{2}$ is rational no.

$$\sqrt{2} = \frac{p}{q} \quad \left(\begin{array}{l} p \text{ \& } q \text{ doesn't} \\ \text{have any common} \\ \text{factor} \end{array} \right)$$

squaring both sides

$$(\sqrt{2})^2 = \left(\frac{p}{q}\right)^2$$

$$2 = \frac{p^2}{q^2}$$

$$\boxed{q^2 = \frac{p^2}{2}} \quad (1)$$

2 divides p^2
then 2 divides p also.

$$(\text{let}) \quad n = \frac{p}{2}$$

$$\boxed{p = 2n}$$

put in (1)

$$q^2 = \frac{(2n)^2}{2}$$

$$q^2 = \frac{4n^2}{2}$$

$$\frac{q^2}{2} = n^2$$

2 divides q^2
then 2 divide q also

$$(\text{let}) \quad m = \frac{q}{2}$$

$$\boxed{q = 2m}$$

$$\frac{p}{q} = \frac{2n}{2m} \rightarrow \text{common factor is } 2$$

This contradiction is due to our wrong assumption.
that $\sqrt{2}$ is rational.

Hence, $\sqrt{2}$ is irrational no.

HP

Axiom

if p divides a^2

then p divide 'a' also.

for eg

$$5 \longrightarrow 25 \text{ i.e. } (5)^2$$

$$5 \longrightarrow 5 \text{ also}$$

#LP: Prove that $\sqrt{5}$ is an irrational number.

Let $\sqrt{5}$ is rational.

$\sqrt{5} = \frac{p}{q}$ (p & q have no factor in common)

sq. both sides

$$5 = \frac{p^2}{q^2}$$

$$\boxed{q^2 = \frac{p^2}{5}} \quad \text{--- (1)}$$

5 divides p^2
5 divide p also

~~(let) $m = \frac{p}{5}$~~

$$\boxed{p = 5m}$$

put in (1)

$$q^2 = \frac{(5m)^2}{5}$$

$$q^2 = \cancel{5} \frac{5m^2}{\cancel{5}}$$

$$\boxed{\frac{q^2}{5} = m^2}$$

5 divides q^2
5 divide q also

[CBSE 2019]

~~(let) $n = \frac{q}{5}$~~

$$\boxed{q = 5n}$$

$$\frac{p}{q} = \frac{5m}{5n} \rightarrow 5 \text{ is common factor}$$

this contradiction is due to our wrong assumpt

Hence, $\sqrt{5}$ is irr. **(HP)**

#(3B)

$\mathbb{R}$

$$\left. \begin{array}{l} \mathbb{R} + \mathbb{R} \\ \mathbb{R} - \mathbb{R} \\ \mathbb{R} \times \mathbb{R} \\ \mathbb{R} / \mathbb{R} \end{array} \right\} \rightarrow \mathbb{Q}$$

$\mathbb{R}, \mathbb{I}\mathbb{R}$

$$\left. \begin{array}{l} \mathbb{R} + \mathbb{I}\mathbb{R} \\ \mathbb{R} - \mathbb{I}\mathbb{R} \\ \mathbb{R} \times \mathbb{I}\mathbb{R} \\ \mathbb{I}\mathbb{R} / \mathbb{R} \end{array} \right\} \rightarrow \mathbb{I}\mathbb{R}$$

$\mathbb{I}\mathbb{R}, \mathbb{I}\mathbb{R}$

$$\left. \begin{array}{l} \mathbb{I}\mathbb{R} + \mathbb{I}\mathbb{R} \\ \mathbb{I}\mathbb{R} - \mathbb{I}\mathbb{R} \\ \mathbb{I}\mathbb{R} \times \mathbb{I}\mathbb{R} \\ \mathbb{I}\mathbb{R} / \mathbb{I}\mathbb{R} \end{array} \right\} \rightarrow \text{Cannot say.}$$

#LP:

Given that $\sqrt{2}$ is irrational, prove that $(5+3\sqrt{2})$ is an irrational number.
(2018)

let $5+3\sqrt{2}$ is rational.

$$5+3\sqrt{2} = \frac{p}{q}$$

$$3\sqrt{2} = \frac{\cancel{p}}{\cancel{q}} - 5 \quad \begin{matrix} \text{R} & \text{R} & \text{R} \end{matrix}$$

$$\sqrt{2} = \frac{\frac{\cancel{p}}{\cancel{q}} - 5}{3} \quad \begin{matrix} \text{R} & \text{R} & \text{R} \end{matrix} \Rightarrow \text{R}$$

$\swarrow$
Irr
(given)

$$\text{Irr} \neq \text{R}$$

This contradiction is
due to our wrong
assumption.

Hence $5+3\sqrt{2}$ is irr.

#LP:

Prove that $\left(5\sqrt{3} + \frac{2}{3}\right)$ is an irrational number given that $\sqrt{3}$ is an irrational number. (2025)

let $5\sqrt{3} + \frac{2}{3}$ is rational

$$5\sqrt{3} + \frac{2}{3} = \frac{p}{q}$$

$$5\sqrt{3} = \frac{p}{q} - \frac{2}{3}$$

$$\sqrt{3} = \frac{\frac{p}{q} - \frac{2}{3}}{5}$$

Ir
Given

$$\overline{\mathbb{R}} \neq \mathbb{R}$$

this contra —

$$\therefore 5\sqrt{3} + \frac{2}{3} \text{ is irrat.}$$

HP

#LP: Prove that $(\sqrt{2} + \sqrt{3})^2$ is an irrational number, given that $\sqrt{6}$ is an irrational number.

Let $(\sqrt{2} + \sqrt{3})^2$ is rational.

$$(\sqrt{2} + \sqrt{3})^2 = \frac{p}{q}$$

$$(\sqrt{2})^2 + (\sqrt{3})^2 + 2(\sqrt{2})(\sqrt{3}) = \frac{p}{q}$$

$$2 + 3 + 2\sqrt{2 \times 3} = \frac{p}{q}$$

$$5 + 2\sqrt{6} = \frac{p}{q}$$

$$2\sqrt{6} = \left[\frac{p}{q} - 5 \right]_{\mathbb{R}}$$

$$\sqrt{6} = \frac{\left[\frac{p}{q} - 5 \right]_{\mathbb{R}}}{2} \rightarrow \text{Irr (given)}$$

$$\text{Irr} \neq \mathbb{R}$$

this contr. - - -

Hence $(\sqrt{2} + \sqrt{3})^2$ is irr.
HP

9/14 Completed!!

Thank you Goodies !!

